Design Project Proposal Template

Fall 2026

Use this template to prepare your 1–2 page proposal for the project. You do not need to write full essays for each item—concise answers are fine. Submit one proposal per group on Canvas.

1 Project Title

Enter a short, descriptive name for your project.

FocusBird: An interactive flappy bird inspired game, using the Grove IO kit for the Arduino

2 Team Members

Group List all members and their main role/contribution.

Angad Virk: Raylib mechanics

Louis Vo: Arduino inputs and outputs.

Zaarin Tasneem: Hardware mapping, game design.

3 Problem Statement

What problem or task will your project address? Who would use it, and why would it be useful or interesting?

FocusBird is an interactive Arduino game inspired by Flappy Bird. Players guide a bird through obstacles using a button, sound sensor, and light sensor, responding to different instructions along the way. Designed for young children, the game makes practicing attention and reaction time more engaging and approachable.

4 Project Objectives

List the main goals. Explain how your project will involve enough design and programming work for your group.

- Create a game that utilizes hardware inputs intended to help children focus
- We will implement child friendly feedback (fun noises or a congratulation)
- Show prompts, response times and gameplay scores separately
- Implement various difficulty levels and maps

This project will involve designing an interactive game that will not only be engaging for the user, but also cater to the hardware of the arduino grove kit where the game will be run. The design work will be making the game utilize inputs and outputs of the arduino grove kit and the gameplay mechanics.

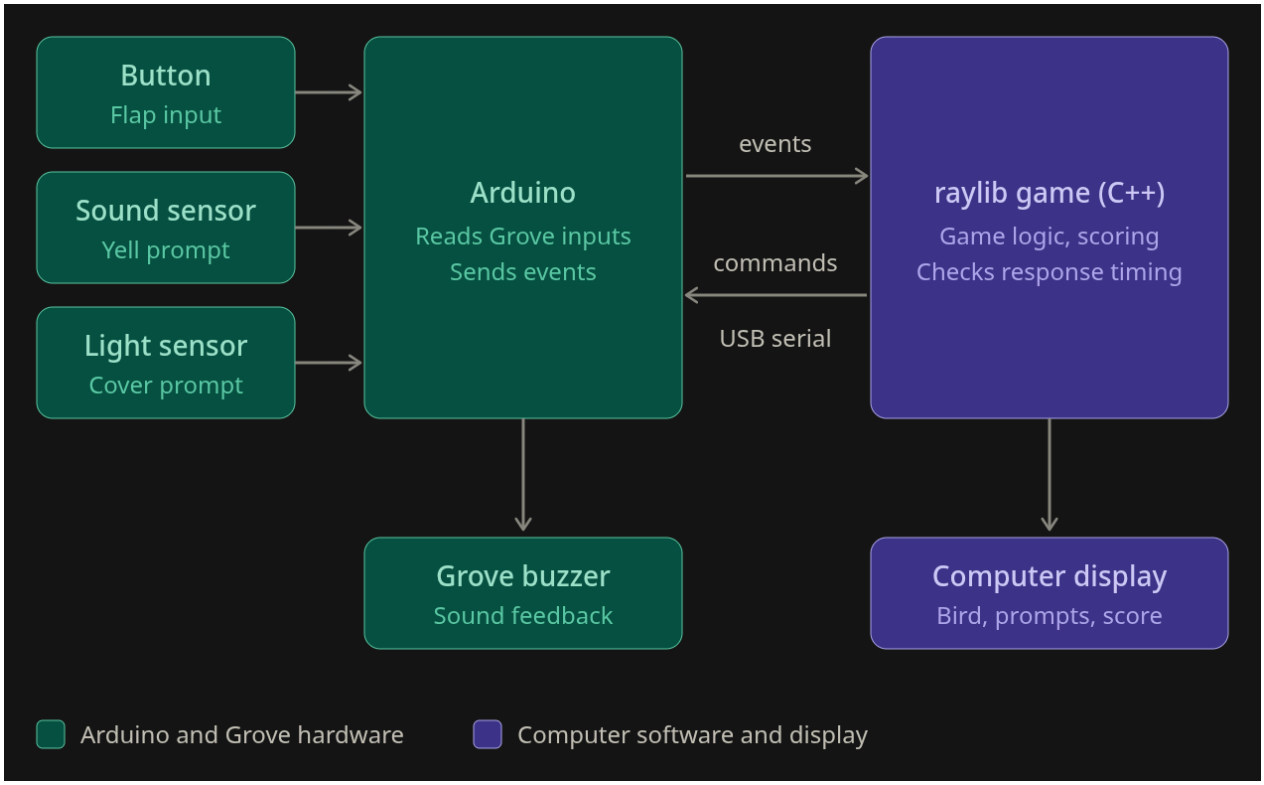
5 Proposed Solution and Initial Design

Describe how someone will use the system, from an input or user action to the resulting output. Explain the main parts of your program and how they work together. You may include a simple sketch or block diagram.

State what will run on the Arduino. If you also use a C++ program on a computer, explain its tasks and what readings or commands will pass between the two programs, for example over USB.

The player controls a bird in a computer game by responding to on-screen prompts. A button makes the bird flap. When an obstacle appears with a certain prompt, the player must act accordingly, interacting with a specific arduino component to interact with the mage. Correct inputs help the bird pass the obstacle, while missed or incorrect inputs lead to the player losing.

The Arduino will read the Grove button, sound sensor, and light sensor, and send their readings or detected events to a C++ game program over USB serial. The game program, built with raylib, will display the bird, obstacles, prompts, score, and feedback. It will send commands to the Arduino when needed, such as triggering the buzzer to signal an upcoming obstacle or provide feedback. The Arduino and game will communicate using a simple, consistent serial message format.



6 Design Specifications

List the main tasks your system must do. Be clear about the inputs, decisions, and outputs. Include settings, timing, or limits where relevant. For example: the user selects a recording duration, and the program saves sensor readings until that time has elapsed.

State any important limits, such as the available components, servo movement range, or the need for a connected computer.

The system must:

- Display a playable side-scrolling game with a bird, obstacles, prompts, and score.
- Use the Grove button as the flap input.
- Detect a sound with the sound sensor when a “Yell” prompt is shown.
- Detect a covered light sensor when a “Cover!” prompt is shown.
- Check whether the player’s response matches the prompt and arrives within the allowed response time.
- Track successful responses, misses, and reaction times, and show the results during or after gameplay.
- Provide child-friendly feedback, such as a buzzer sound or on-screen congratulatory message.
- Support at least two difficulty levels, with differences such as obstacle spacing or response time.

The game will run on a computer using raylib, and the Arduino must be connected by USB while playing. Sensor thresholds and response times will be set during testing.

7 Tools and Resources

List the Arduino/Grove components, software, and libraries you plan to use. For each sensor or output component, briefly state its purpose.

You may use your own sensors or other components after checking with a TA and obtaining approval before use.

- Sound sensor
- Button
- Light sensors
- Buzzer
- Raylib library.
- C++ Standard Library
- Wire library
- etc…

8 Expected Results and Demonstration

Describe what you expect the completed project to do. Explain what you will show the TA and how you will check the results. Include a normal operating example and a relevant special case, such as a held button or a reading near a threshold.

We will show the bird flapping with the button, a sound prompt triggering before a sound, and a light prompt triggering before the light sensor is covered. The bird will get past obstacles each time it receives an input. We will show the TA the above gameplay and we will check the result through correct/misses, reaction times.
Normal: YELL appears, a sound is detected within the window, obstacle is avoided.
Special case: covering the light sensor before the prompt must not count

*Note that specific prompts may be subject to change but the idea remains the same

9 Potential Challenges

Identify the main difficulties you expect, such as choosing sensor thresholds, coordinating

timed actions, or exchanging data with a computer. State how you will investigate them and what simpler approach you could use if needed.

Some difficulties that we expect is finding the right threshold for the sound sensor in order to play it in different environments, glitches of the bird phasing through obstacles or overlapping (2d body collision), data shown in wrong locations.

We expect challenges with tuning the sound and light sensor thresholds, preventing repeated triggers, and coordinating sensor inputs with timed game prompts over USB serial; we will investigate these by testing each component separately, then testing the integrated system under different conditions. If detection or communication proves unreliable, we will use simpler threshold-based inputs with a cooldown and a basic, clearly formatted serial message for each detected action.

10 Timeline and Milestones

Table plan for design, coding, testing, and demos

Date	Timeline
Oct 20	Get Github repo setup, decide what libraries to use
Oct 27	Make basic code run with flappy bird game, create design for finished game
Nov 3	Figure out specific arduino inputs and map them with game
Nov 10	Showcase project up to that point, continue mapping inputs
Nov 13	Project Review Report due
Nov 17	Finalize code and hardware design, address any problems
Nov 24	Demonstrate final Project!
Nov 30	Final Submission for project